INTEGRATING PROMETHEUS WITH GRAFANA FOR MONITORING

Micro Project Report

Submitted by

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BONAFIDE CERTIFICATE

Certified that this micro project documentation of "**INTEGRATING PROMETHEUS WITH GRAFANA FOR MONITORING"** is the bonafide work of **"RANJANI K S (22CDR084), VIMALANANDHAN S (21CDR058)"** who carried out the project under my supervision. Certified further that to the best of my knowledge the work reported here in does not form part of my other thesis or dissertation on the basis of which a degree or awarded was conferred on an earlier occasion on this or any other candidate.

Submitted on 07.05.25

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INTEGRATING PROMETHEUS WITH GRAFANA FOR MONITORING

AIM

The aim of this project is to develop a reliable and real-time monitoring system by integrating Prometheus with Grafana. It focuses on collecting system metrics such as CPU, memory, and disk usage from an Ubuntu virtual machine using Node Exporter. The collected data is visualized using Grafana dashboards to provide meaningful insights. The setup ensures continuous monitoring by using Systemd to run Prometheus and Node Exporter in the background. Nano is used as a simple text editor to modify these configuration files easily.

ABSTRACT

Monitoring the performance of systems is crucial for maintaining uptime and detecting issues proactively. This project demonstrates the integration of Prometheus and Grafana to build a powerful monitoring solution. Prometheus is used to collect and store time-series metrics from an Ubuntu virtual machine, with Node Exporter providing detailed system-level data. Grafana connects to Prometheus and transforms this data into visually rich and interactive dashboards. The services are managed using Systemd to ensure they run continuously in the background. Configuration is performed through YAML files, offering a clear and manageable setup structure. Nano is used for quick and efficient editing of these files. The entire monitoring stack runs on an Ubuntu VM, making it portable and easy to deploy. This solution allows for real-time analysis and visualization of system health. Overall, the integration enhances the ability to monitor, analyze, and troubleshoot system performance efficiently.

SCOPE AND OBJECTIVES

SCOPE:

- Focuses on setting up a real-time system monitoring solution using Prometheus, Node Exporter, and Grafana on an Ubuntu virtual machine.
- Covers the complete process of collecting system metrics, configuring Prometheus, and displaying data through Grafana dashboards.
- Demonstrates how to track critical system metrics such as CPU usage, memory consumption, and disk space in real time.
- Reinforces the practical understanding of how system monitoring tools can be integrated for better visibility and performance tracking.
- Emphasizes data visualization, continuous system health monitoring, and ease of setup using lightweight tools and configurations.

OBJECTIVES :

- To install and configure Prometheus for collecting system-level metrics such as CPU, memory, and disk usage.
- To integrate Node Exporter with Prometheus to expose system metrics from the Ubuntu VM.
- To visualize the collected data using Grafana dashboards with user-friendly charts and graphs.
- To ensure Prometheus and Node Exporter run continuously in the background using Systemd.
- To manage Prometheus configuration using YAML files for clarity and reusability.
- To edit configuration files efficiently using the Nano text editor.

HARDWARE AND SOFTWARE REQUIREMENT

Hardware Requirements:

- Internet connection to install tools and send alerts
- Minimum 4 GB RAM (8 GB recommended for better performance)
- At least 1 GB of free storage space
- A screen size of 13" or larger with at least 1366×768 resolution

Software Requirements:

- **Ubuntu Virtual Machine** – To host and run Prometheus, Node Exporter, and Grafana.
- **Prometheus** – For collecting and storing system metrics.
- **Grafana** – For visualizing the metrics collected by Prometheus.
- **Node Exporter** – To expose system-level metrics (CPU, memory, disk usage).
- **Systemd** – For managing Prometheus and Node Exporter as background services.
- **Nano (or any text editor)** – For editing configuration files (YAML format).
- **Web Browser (Chrome, Firefox, or Edge)** – To access and interact with the Grafana dashboard interface.

PROJECT DESIGN

- The project sets up a monitoring environment on an Ubuntu virtual machine (VM) using Prometheus and Grafana.
- Prometheus is configured to scrape metrics exposed by Node Exporter installed on the same VM.
- Configuration files are written in YAML format for structured and readable setup.
- Grafana is connected to Prometheus as a data source and used to create customizable dashboards.
- The setup is lightweight, efficient, and does not consume excessive system resources.
- Nano or any terminal-based text editor is used to modify configuration files directly within the VM.
- The final outcome is a fully functional, real-time system monitoring solution that can be extended or adapted for larger infrastructure.

ADVANTAGES

- Real-time monitoring of CPU, memory, and disk.
- Visual dashboards in Grafana.
- Option to add alerts for critical metrics.
- Open-source and free tools.
- Runs efficiently in the background.
- Easy setup on local or cloud systems.
- Scalable and customizable.
- Helps prevent system failures early.

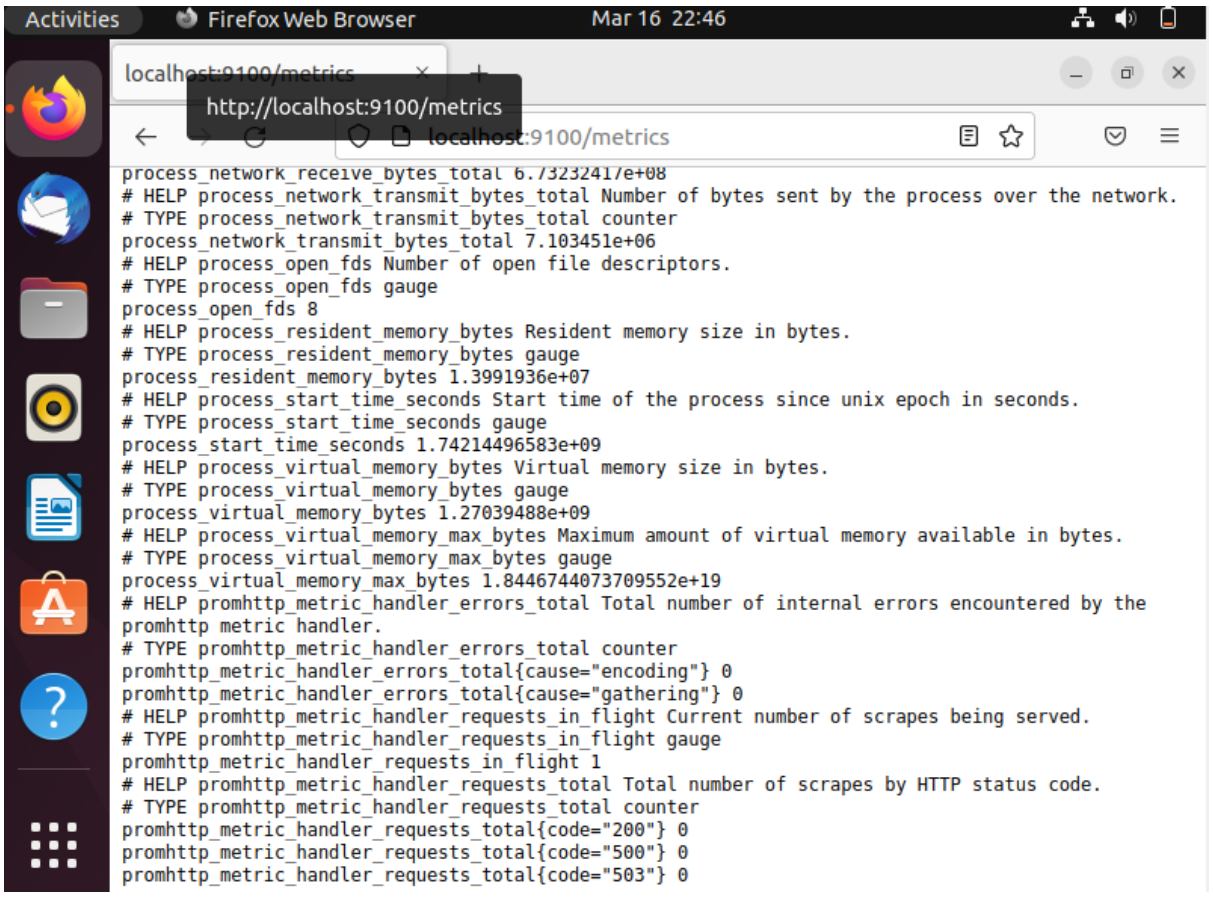
PROPOSED SYSTEM

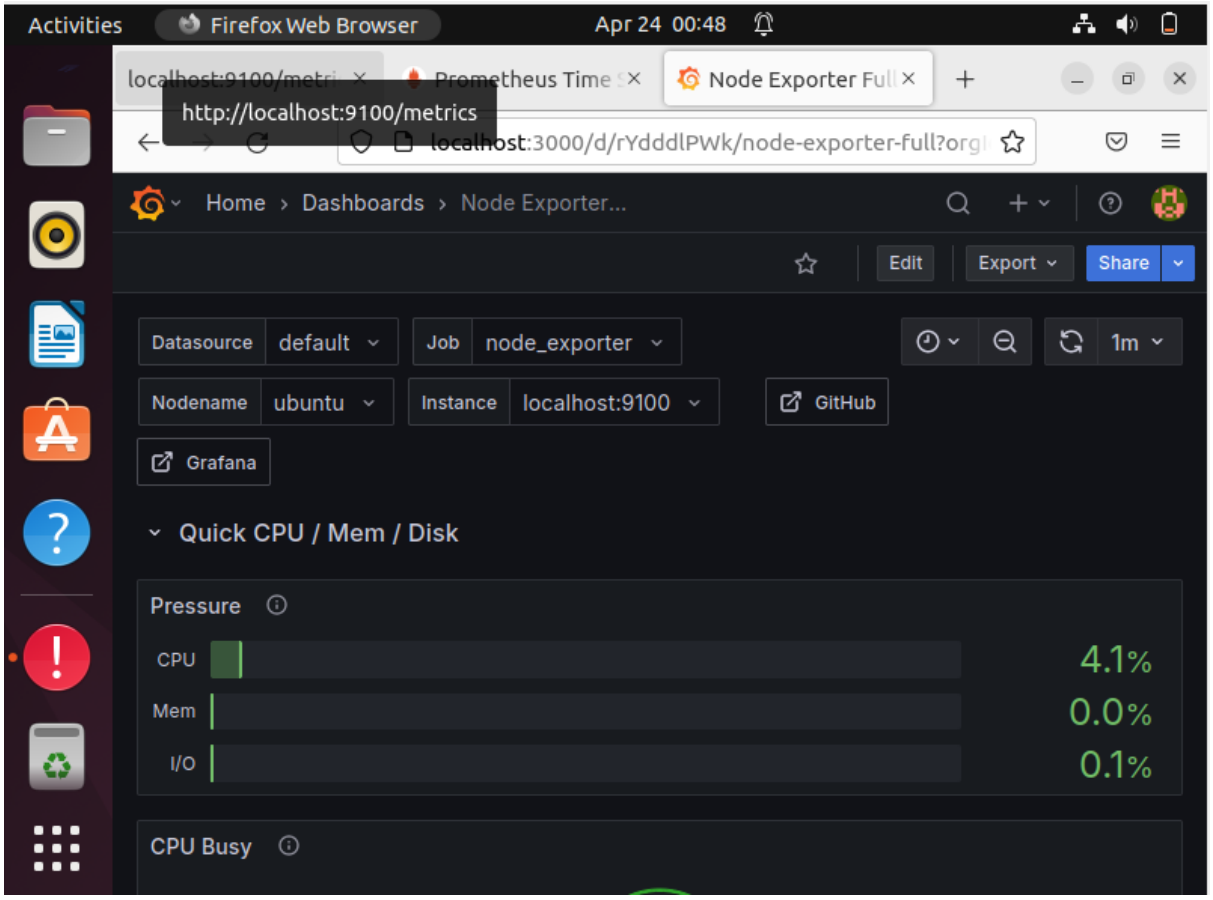
- Uses **Prometheus** to collect and store time-series system metrics from an Ubuntu-based virtual machine.
- Implements **Node Exporter** to expose system-level metrics like CPU, memory, and disk usage to Prometheus.
- Uses **YAML configuration files** to define Prometheus scraping targets and settings, enabling easy and repeatable setup.
- Employs **Systemd** to run Prometheus and Node Exporter as background services for continuous monitoring.
- Integrates **Grafana** as a visualization tool to create user-friendly and customizable dashboards.
- Supports **real-time monitoring** and scalable design, allowing the system to be extended to cloud or multi-node environments.

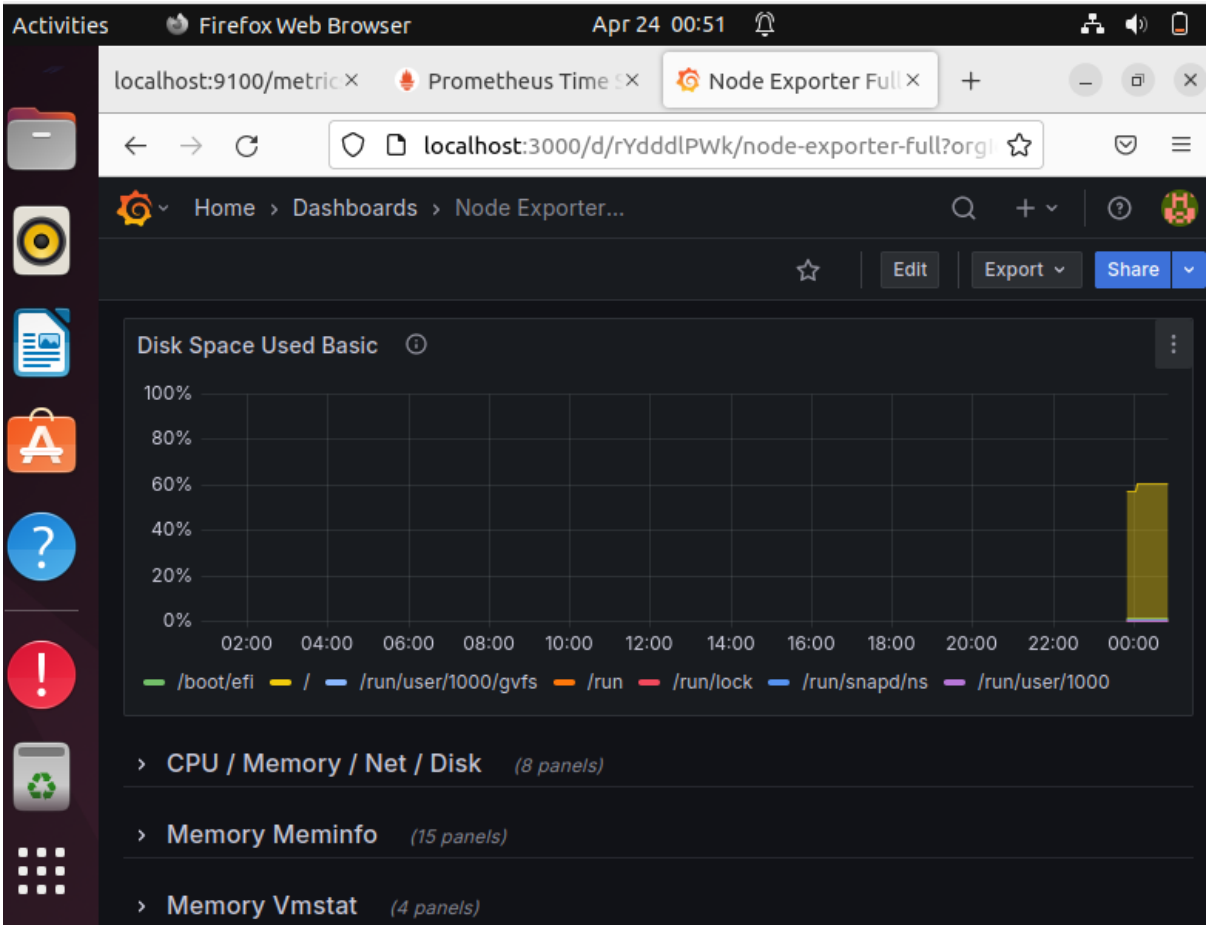
IMPLEMENTATION OF THE PROJECT

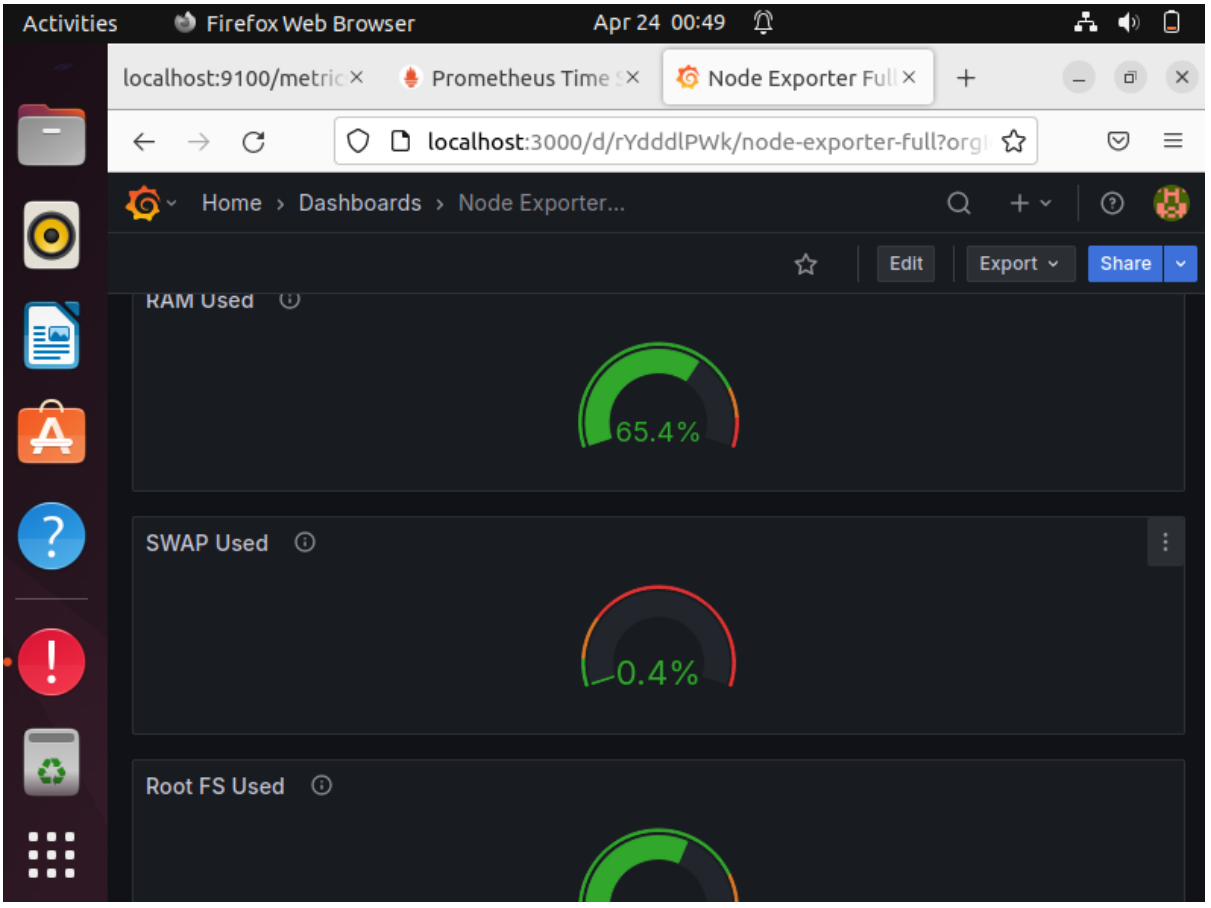
The project was implemented by first setting up an Ubuntu virtual machine as the host environment. Prometheus was installed and configured using YAML files to scrape metrics from Node Exporter, which was also installed on the same system. Node Exporter provided essential system data like CPU, memory, and disk usage. Systemd service files were created to ensure Prometheus and Node Exporter ran automatically in the background. Grafana was then installed and linked to Prometheus as a data source. Dashboards were created in Grafana to visualize the metrics in real time. The entire configuration was tested and refined using simple tools like Nano to edit files. The result was a lightweight, secure, and fully functional monitoring system capable of tracking system health efficiently and in real time.

OUTPUT


Activities
Firefox Web Browser
Mar 16 22:46
localhost:9100/metrics
http://localhost:9100/metrics
localhost:9100/metrics
process_network_receive_bytes_total 6.7323217e+08
HELP process_network_transmit_bytes_total Number of bytes sent by the process over the network.
TYPE process_network_transmit_bytes_total counter
process_network_transmit_bytes_total 7.103451e+06
HELP process_open_fds Number of open file descriptors.
TYPE process_open_fds gauge
process_open_fds 8
HELP process_resident_memory_bytes Resident memory size in bytes.
TYPE process_resident_memory_bytes gauge
process_resident_memory_bytes 1.3991936e+07
HELP process_start_time_seconds Start time of the process since unix epoch in seconds.
TYPE process_start_time_seconds gauge
process_start_time_seconds 1.74214496583e+09
HELP process_virtual_memory_bytes Virtual memory size in bytes.
TYPE process_virtual_memory_bytes gauge
process_virtual_memory_bytes 1.27039488e+09
HELP process_virtual_memory_max_bytes Maximum amount of virtual memory available in bytes.
TYPE process_virtual_memory_max_bytes gauge
process_virtual_memory_max_bytes 1.8446744073709552e+19
HELP promhttp_metric_handler_errors_total Total number of internal errors encountered by the
promhttp metric handler.
TYPE promhttp_metric_handler_errors_total counter
promhttp_metric_handler_errors_total{cause="encoding"} 0
promhttp_metric_handler_errors_total{cause="gathering"} 0
HELP promhttp_metric_handler_requests_in_flight Current number of scrapes being served.
TYPE promhttp_metric_handler_requests_in_flight gauge
promhttp_metric_handler_requests_in_flight 1
HELP promhttp_metric_handler_requests_total Total number of scrapes by HTTP status code.
TYPE promhttp_metric_handler_requests_total counter
promhttp_metric_handler_requests_total{code="200"} 0
promhttp_metric_handler_requests_total{code="500"} 0
promhttp_metric_handler_requests_total{code="503"} 0


Activities
Firefox Web Browser
Apr 24 00:48
Prometheus Time
Node Exporter Full
http://localhost:9100/metrics
localhost:3000/d/rYdddlPWk/node-exporter-full?orgI
Home › Dashboards › Node Exporter...
Edit
Export
Share
Datasource
default
Job
node_exporter
1m
Nodename
ubuntu
Instance
localhost:9100
GitHub
Grafana
Quick CPU / Mem / Disk
Pressure
CPU
4.1%
Mem
0.0%
I/O
0.1%
CPU Busy


Activities
Firefox Web Browser
Apr 24 00:51
localhost:9100/metric
Prometheus Time
Node Exporter Full
localhost:3000/d/rYdddlPWk/node-exporter-full?orgI
Home › Dashboards › Node Exporter...
Edit
Export
Share
Disk Space Used Basic
100%
80%
60%
40%
20%
0%
02:00
04:00
06:00
08:00
10:00
12:00
14:00
16:00
18:00
20:00
22:00
00:00
/boot/efi
/
/run/user/1000/gvfs
/run
/run/lock
/run/snapd/ns
/run/user/1000
CPU / Memory / Net / Disk (8 panels)
Memory Meminfo (15 panels)
Memory Vmstat (4 panels)


Activities
Firefox Web Browser
Apr 24 00:49
localhost:9100/metric
Prometheus Time
Node Exporter Full
localhost:3000/d/rYdddlPWk/node-exporter-full?orgI
Home › Dashboards › Node Exporter...
Edit
Export
Share
RAM Used
65.4%
SWAP Used
0.4%
Root FS Used

CONCLUSION

Integrating Prometheus and Grafana offers a powerful monitoring solution for system performance analysis. Prometheus collects time-series data, while Grafana provides intuitive and interactive dashboards for visualization. Using Node Exporter ensures detailed system-level metrics are captured on an Ubuntu virtual machine. Systemd helps manage the services, ensuring continuous background operation. The setup is simplified with YAML configuration files and quick editing via Nano. This solution empowers real-time monitoring and efficient troubleshooting, enhancing system uptime and performance management.

COMMANDS

```
# 1. Install Prometheus, Grafana, and Node Exporter Manually

# Install Prometheus on Ubuntu
sudo apt update
sudo apt install prometheus

# Start Prometheus
sudo systemctl start prometheus
sudo systemctl enable prometheus

# Install Grafana on Ubuntu
sudo apt-get install -y software-properties-common
sudo add-apt-repository "deb https://packages.grafana.com/oss/deb stable main"
sudo apt-get update
sudo apt-get install grafana
```

```
# Start Grafana
sudo systemctl start grafana-server
sudo systemctl enable grafana-server

# Download and install Node Exporter
tar -xvf node_exporter-*.*.*.tar.gz
sudo mv node_exporter-*/node_exporter /usr/local/bin/

# Run Node Exporter
nohup /usr/local/bin/node_exporter &

# 2. Access Prometheus, Grafana, and Node Exporter

# Prometheus: Open http://localhost:9090
# Grafana: Open http://localhost:3000 (Login: admin / admin)
# Node Exporter: Open http://localhost:9100/metrics

# 3. Configure Prometheus to Scrape Node Exporter

# Edit Prometheus config
sudo nano /etc/prometheus/prometheus.yml

# Add Node Exporter scraping config
# scrape_configs:
#   - job_name: 'node_exporter'
#     static_configs:
#       - targets: ['localhost:9100']
```

# Restart Prometheus
sudo systemctl restart prometheus

4. Set Up Grafana Dashboard

Log in to Grafana (http://localhost:3000) with admin/admin
Add Prometheus as a data source: Go to Configuration → Data Sources → Add Data Source → Prometheus
Set URL as http://localhost:9090 and save